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Project Report - Phase 1

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CHAPTER 1

INTRODUCTION

Music profoundly shapes and reflects human emotions, making it an essential component of personal expression and daily experience. In today's digital era, streaming platforms provide access to vast libraries of songs, yet most recommendation systems still rely heavily on user history, predefined playlists, or collaborative filtering techniques. Such approaches often fail to consider the listener's immediate emotional state, environmental context, or surrounding conditions, leading to suggestions that feel generic or disconnected from the moment. Muzly bridges this gap by offering an AI-driven, context-aware music recommendation system capable of adapting in real time. By integrating computer vision for facial emotion detection with contextual intelligence derived from scene analysis, weather conditions, and location awareness, Muzly generates music recommendations that genuinely resonate with the user's present mood and environment. This results in a deeply immersive, personalized, and emotionally aligned listening experience.

1.1 Objective

- To develop a robust facial emotion recognition module powered by a CNN trained on the FER-2013 dataset, enabling accurate identification of emotional states.
- To improve contextual awareness by leveraging the CLIP (Contrastive Language–Image Pre-training) model for scene analysis and extraction of weather-related cues from user images.

- To integrate real-time contextual inputs—such as geographic location and live climate data—via external APIs to provide continuously adaptive music recommendations.
- To create a weighted, context-sensitive recommendation system that seamlessly combines emotional cues, scene details, location insights, and environmental factors to generate highly personalized music suggestions.

1.2 Overview

Muzly is a sophisticated multimodal music recommendation platform powered by AI, combining computer vision, deep learning, and contextual analysis to generate highly personalized song suggestions. It evaluates the user's facial expressions—taken from an uploaded photo or live camera feed—and uses a CNN trained on annotated emotion datasets to detect feelings such as joy, sadness, anger, or surprise.In parallel, the CLIP (Contrastive Language–Image Pre-training) model examines the same image to interpret the surrounding scene and infer climatic conditions, providing deeper environmental context. These visual insights are further enriched with real-time location and weather information retrieved through external APIs. All extracted parameters are processed by a weighted recommendation engine that intelligently combines emotional cues, scene understanding, climatic factors, and language preferences to suggest suitable songs from a multilingual database. The entire system is delivered through a modern, responsive user interface built using React, offering users a smooth, interactive, and visually engaging experience. This frontend integrates seamlessly with the backend services, enabling real-time analysis and dynamic music recommendations supported by clear, intuitive visual feedback.

1.3 Problem Statement

Modern music recommendation systems predominantly rely on statistical modeling, user listening history, and collaborative filtering techniques to generate song suggestions. Although these approaches can produce acceptable results, they overlook a crucial dimension of user experience—the listener's immediate emotional state and environmental context. Consequently, users often encounter recommendations that feel repetitive, irrelevant, or disconnected from the moment. This gap reveals the limitations of traditional recommendation strategies, emphasizing the need for an adaptive, emotionally intelligent system capable of responding dynamically to real-time user conditions. Muzly aims to address this challenge by incorporating emotion recognition, scene understanding, and contextual awareness to provide music suggestions that truly resonate with the user's current mood and surroundings.

1.4 Motivation

Human emotions are fluid and constantly shaped by internal feelings as well as external influences such as weather, environment, social setting, and personal experiences. Traditional music recommendation systems fail to capture this dynamic nature, often offering generic playlists that do not align with the listener's immediate emotional or contextual needs. The motivation behind **Muzly** arises from the vision to bridge this disconnect by creating an intelligent, context-aware music recommendation platform that integrates real-time emotion detection and environmental analysis. By tailoring music suggestions to reflect the listener's current mood, scene, and surrounding conditions, Muzly aspires to deliver a more meaningful, immersive, and deeply personalized listening experience. This ensures that every recommendation feels intentional, relevant, and emotionally aligned with the user's real-time state.

4

1.5 Scope of the Project

- Real-time recognition of user emotions through computer vision and deep learning methods, enabling accurate analysis of facial expressions from uploaded photos or live camera feeds.
- Scene and climatic context analysis through AI-driven image interpretation using models such as CLIP to understand the user's environment and surrounding conditions.
- Integration of geographical and weather data obtained through external APIs to further refine and contextualize music recommendations.
- A modern, responsive, and interactive front-end interface built with React, ensuring accessibility, smooth user interactions, and real-time feedback.
- A curated, multilingual song database designed to accommodate diverse cultural and linguistic preferences.
- A weighted recommendation engine that intelligently combines emotional cues, scene context, language preferences, and climate factors to generate highly accurate and context-aware music suggestions.
- Efficient preprocessing and synchronization of multimodal inputs to ensure reliable, fast, and seamless system performance across different scenarios.
- Modular and scalable system architecture that allows easy integration of future models, additional emotion classes, or extended data sources without major redesign.
- Secure handling of user images and data, ensuring privacy through temporary processing, non-storage policy, and safe API communication practices.
- High-performance backend implementation using FastAPI, providing rapid request handling, low latency music recommendations, and smooth

1.6 Direct Benefits

- Delivers emotionally relevant and context-aware music recommendations, ensuring that songs align with the listener's current mood and environment.
- Enhances user engagement by dynamically adapting song suggestions to real-time emotional and contextual inputs.
- Provides a seamless and intelligent user interface capable of processing inputs instantly through a responsive React frontend.
- Adapts automatically to fluctuations in emotion, surroundings, and climatic conditions, offering consistently relevant music suggestions.
- Reduces user effort by eliminating the need for manual searches or playlist curation.
- Improves the overall listening experience through meaningful, personalized, and situation-aware music recommendations.
- Increases system usability and accessibility, as users can interact effortlessly with the interface without requiring prior knowledge or manual settings.

1.7 Indirect Benefits

- Encourages research and innovation in multimodal artificial intelligence and human–computer interaction.
- Provides potential utility in mental wellness, therapeutic contexts, and interactive advertising, where emotion-aware systems are beneficial.
- Supports the development of adaptive, emotion-responsive interfaces, paving the way for smarter AI-driven platforms.
- Offers valuable insights for future studies in affective computing, context-aware systems, and emotion-based personalization.

- Demonstrates how AI-powered personalization can elevate digital media consumption, leading to richer and more meaningful user experiences.
- Contributes to advancements in user-centric design, highlighting the importance of emotion-driven interaction models.
- Promotes ethical and responsible AI development, particularly in areas of privacy-conscious image processing and contextual data handling.

1.6 Organization of the Report

This project report is organized to present a clear, coherent, and in-depth overview of the design, development, and implementation of the Muzly system. Each chapter is organized to present the progression of ideas from conceptual foundations to practical execution and evaluation.

14

Chapter 1	Introduction
Chapter 2	Literature Review
Chapter 3	System Implementation
Chapter 4	Source Code
Chapter 5	Results and Evaluation
Chapter 6	Conclusion and Future Works

CHAPTER 2
LITERATURE REVIEW

The field of emotion-based music recommendation has witnessed significant advancements in recent years, largely propelled by advancements in deep learning technologies and multimodal emotion recognition techniques. Traditional recommendation systems primarily relied on user behavior or explicit feedback, often overlooking the dynamic emotional state of the listener. Recent studies, however, emphasize understanding users' emotions through various modalities including facial expressions, speech, EEG signals, and textual data to deliver personalized music experiences. Researchers have explored diverse approaches, from LSTM and CNN-based models to hybrid and context-aware systems, highlighting the importance of combining temporal, visual, and auditory cues for more accurate emotion prediction. This body of work collectively demonstrates a trend towards highly adaptive, user-centric music recommendation systems that respond in real-time to emotional states, offering an enhanced and engaging listening experience.

2.1. Emotion-Based Music Recommendation Using Deep Learning Models

Deep learning approaches have revolutionized emotion-based music recommendation systems by enabling the modeling of complex and nonlinear patterns in user data. Joel et al. [1] developed a deep neural network model that predicts a user's emotional state from input features and recommends music tracks that align with that state. Their model incorporates multiple layers to capture subtle patterns in emotional expression, demonstrating improved personalization over traditional content-based methods. Joshi et al. [2] focused on temporal modeling using LSTM networks, which capture sequential dependencies in emotional data, allowing the system to adapt music recommendations as users' moods change over

time. Katkuri et al. [3] implemented a hybrid CNN-SVM framework: the CNN automatically extracts deep emotional features from inputs such as images or audio signals, while the SVM performs classification to map these features to discrete emotional categories. Sharma and Bansal [9] designed a system capable of real-time adaptation, showing that deep learning models can dynamically adjust recommendations based on live emotion detection, thus enhancing user engagement. Collectively, these studies demonstrate that deep learning-based architectures provide robust solutions for mapping complex emotional patterns to personalized music recommendations.

2.2. Multimodal Emotion Recognition for Enhanced Recommendations

To improve recommendation accuracy, researchers have increasingly explored multimodal emotion detection, combining cues from multiple modalities input channels. Maji et al. [4] introduced a deep temporal fusion network that integrates facial expressions, speech signals, and physiological measurements, such as heart rate and skin conductance, to achieve a more accurate understanding of user emotions. Pan et al. [5] extended this idea by combining facial, speech, and EEG modalities, emphasizing the complementary nature of these signals for detecting nuanced emotional states. Tripathi et al. [6] leveraged the IEMOCAP dataset, which contains speech, text, and facial recordings, to evaluate multimodal emotion recognition models, highlighting the benefits of combining multiple inputs over single-modal approaches. Priyasad et al. [16] employed an attention-driven fusion mechanism to selectively weigh information from different modalities, demonstrating that attention-based methods can enhance the performance of emotion recognition systems by focusing on the most informative signals. These studies collectively underline that multimodal inputs not only strengthen the effectiveness of emotion classification but also provide a richer context for generating personalized music recommendations.

2.3. Facial Emotion-Based Music Recommendation Systems

Facial emotion recognition has emerged as a practical and intuitive method for real-time emotion detection in music recommendation systems. Vayadande et al. [7] and Kuldeep Vayadande et al. [19] proposed frameworks that analyze facial expressions captured via webcam or camera devices to determine the user's emotional state and select appropriate music tracks accordingly. Zacharias et al. [8] developed an image-based system that extracts deep facial features using convolutional neural networks to infer emotion, further demonstrating the feasibility of visual inputs in enhancing user-centric recommendations. These studies show that facial analysis can serve as a reliable and non-intrusive modality for detecting emotions in real-time, providing immediate feedback for adaptive music recommendation systems. The effectiveness of these methods highlights the value of combining computer vision techniques with recommendation systems.

2.4. Hybrid and Context-Aware Music Recommendation Systems

Hybrid and context-aware systems leverage multiple types of data—including content features, user context, and emotional cues—to improve recommendation quality. Wang et al. [10] proposed the CAME (Content- and Context-Aware Music Embedding) model, which fuses information about music content with user context, such as listening history and time-of-day, to deliver more personalized recommendations. Zheng et al. [13] designed a hybrid deep learning system that integrates multimodal emotional data and content features to produce emotion-aware recommendations. Zhou et al. [14] implemented semantic sequence modeling to understand the contextual progression of user interactions, thereby enabling predictive recommendations that anticipate user preferences. Shiralaskar et al. [15] explored AI-driven adaptive systems that move beyond static playlists, emphasizing continuous adjustment of recommendations based on real-time emotional states.

2.5. Recent Innovations and Future Directions

Recent research emphasizes AI-driven and multimodal recommendation systems with enhanced personalization capabilities. Zhang et al. [11] explored neural network architectures for AI-driven emotion-based music recommendations, showing the potential of deep learning in handling large-scale user data and complex emotional patterns. Melchiorre et al. [17] introduced the JAM system, which combines multimodal emotion recognition with natural language processing, allowing users to interact with the system through text or voice queries, resulting in highly personalized music recommendations. Maji et al. [24] and Pan et al. [25] further validated the effectiveness of deep temporal fusion networks for multimodal emotion recognition, demonstrating superior performance over traditional single-modal systems. Collectively, these studies reveal a clear trend towards adaptive, real-time, and emotion-aware recommendation systems that integrate facial, speech, EEG, and behavioral data. This evolution reflects a shift from conventional content-based approaches to sophisticated, user-centric solutions capable of responding dynamically to the listener's emotional state [1–25].

2.6 RELATED BACKGROUNDS

No	Paper Title	Pros	Cons	Research Gap
1	Joel J.S., Thompson B.E., Kumar T.R. (2023) – Emotion-Based Music Recommendation System using Deep Learning Model	Introduced deep learning for emotion-driven recommendations; improved personalization accuracy.	Limited to single emotion modality; lacks contextual awareness.	Integration of multimodal data (scene, weather, location) for richer personalizatio n
2	Joshi S. et al. (2021) – Emotion-Based Music Recommendation Using LSTM	Captures temporal emotion patterns; supports adaptive recommendations.	Computationall y intensive; limited dataset diversity.	Incorporating real-time emotional feedback with optimized LSTM variants.
3	Katkuri S. et al. (2023) – Emotion-Based Music Recommendation Using CNN and SVM	Combined CNN feature extraction with SVM classification for precision.	Lacks dynamic adaptation; fixed input structure.	Hybrid deep models for continuous emotional monitoring.
4	Maji B. et al. (2023) – Multimodal Emotion Recognition with Deep Temporal Fusion	Effectively fuses temporal and multimodal features for emotion detection.	High model complexity; resource-demanding.	Lightweight multimodal architectures for real-time deployment.
5	Pan J. et al. (2023) – Multimodal Emotion Recognition using Facial, Speech, and EEG	High emotion recognition accuracy; diverse input sources.	EEG acquisition impractical for general users.	Replace physiological signals with contextual cues (weather, scene).
6	Tripathi S. et al. (2018) – Multimodal Emotion Recognition on IEMOCAP Dataset	Benchmarked multimodal deep learning methods; comprehensive dataset.	Dataset limited to controlled lab settings.	Real-world multimodal datasets for natural emotional variation.
7	Vayadande K. et al. (2023) – Facial Emotion-Based Song Recommendation	Real-time face-based emotion detection and music mapping.	Depends heavily on lighting and camera quality.	Robust emotion detection under varying conditions.

8	Zacharias V. et al. (2023) – Image-Based Emotion Recognition for Music Recommendation	Innovative use of image-based context for personalization.	Does not account for external factors like location or weather.	Combine image emotion with contextual metadata.
9	Sharma V.P., Bansal A. (2021) – Emotion-Based Music Recommendation System	Demonstrates dynamic emotion recognition for playlist adaptation.	Limited dataset and song variety.	Broader music dataset integration for genre diversity.
10	Wang D. et al. (2021) – Content- and Context-Aware Music Embedding (CAME)	Introduced joint embedding of content and context features.	Lacks emotional inference; focuses only on context.	Emotion-driven content-context hybrid embeddings.
11	Zhang J. et al. (2025) – AI-Driven Emotion-Based Music Recommendation	Advanced AI-driven personalization models with scalable architecture.	Requires high computational resources.	Efficient deep models for mobile and web scalability.
12	Zhang L. et al. (2023) – Multimodal Emotion Recognition with Speech and Vision	High accuracy via bimodal fusion.	Limited emotion categories.	Expand to complex, compound emotion recognition.
13	Zheng Y. et al. (2024) – Hybrid Deep Learning for Emotion-Aware Music Recommender	Integrates multiple modalities for emotion-aware output.	Computational latency; lacks personalization.	Online learning mechanisms for user-specific tuning.
14	Zhou Y. et al. (2022) – Context-Aware Recommendation via Semantic Sequence Modeling	Uses user interaction sequences to improve recommendations.	Does not capture real-time emotion changes.	Integrating sequential modeling with emotion recognition.
15	Shiralaskar H. et al. (2025) – Beyond Playlists: AI-Driven Emotion Recommendation	Dynamic recommendation using adaptive AI logic.	No integration of multimodal emotion detection.	Combine adaptive AI with multimodal context recognition.
16	Priyasad D. et al. (2020) – Attention Driven	Attention mechanism	High training complexity;	Lightweight attention

	Fusion for Multimodal Emotion Recognition	improves cross-modal weighting.	prone to overfitting.	fusion architectures.
17	Melchiorre A.B. et al. (2025) – JAM: Multimodal and Personalized Music Recommendation	Integrates NLP and multimodal emotion recognition for personalized experience.	Still under prototype stage; limited scalability tests.	Scalable deployment with efficient model compression.
18	Lozano-Murciego Á. et al. (2021) – Context-Aware Music Recommender Systems Review	Comprehensive literature synthesis on context-aware systems.	No focus on emotional intelligence.	Extend review to emotional-context-aware systems.
19	Kuldeep Vayadande et al. (2023) – Facial Emotion-Based Song Recommendation	Simple and effective emotion-to-music mapping.	Limited accuracy in complex emotions.	Deep emotion classification for nuanced feelings.
20	Sharma V.P. (2021) – Emotion-Based Music Recommendation System	Early prototype for emotion-based playlist automation.	Static architecture; lacks AI-driven adaptation.	Dynamic feedback integration using real-time emotion updates.
21	Iyer P., Pasad R. (2022) – Emotion-Based Mood Enhancing Music Recommendation	Focuses on mood enhancement for mental health.	Limited scope to emotional wellness, not personalization.	Expanding to multi-purpose adaptive music platforms.
22	Joshi S. et al. (2021) – Emotion-Based Music Recommendation Using LSTM	Models emotional state transitions effectively.	Trained on small emotion datasets.	Utilize larger emotion datasets with multimodal input.
23	Katkuri S. et al. (2023) – CNN + SVM Hybrid for Music Recommendation	Good balance of accuracy and efficiency.	Poor generalization to unseen emotional inputs.	Train on diverse real-world emotional datasets.
24	Maji B. et al. (2023) – Deep Temporal Fusion Networks	Fusion across time for multimodal accuracy.	High training cost and inference time.	Reduce model size via pruning .

CHAPTER 3
SYSTEM IMPLEMENTATION

3.1 Overview

The proposed system is an Emotion- and Scene-Based Music Recommendation Platform designed to intelligently suggest songs according to the user's facial expressions, environmental context, and current weather conditions. The primary objective is to elevate the user's listening experience by generating music recommendations that align with the listener's real-time mood and surroundings, rather than depending solely on historical preferences or manual song selection. To achieve this, the system integrates multiple artificial intelligence models and contextual data sources into a unified workflow. First, the user's facial emotion is detected using a Convolutional Neural Network that has been trained on standard emotion-recognition datasets. In parallel, the CLIP (Contrastive Language–Image Pre-training) model analyzes the uploaded image to infer the surrounding scene—such as home, outdoor, party,and identifies visual cues related to climatic conditions.Additional context is incorporated by retrieving the user's real-time location and weather information through external APIs such as OpenWeatherMap and IPInfo. These multimodal inputs -emotion, scene context, and climate—are processed by a weighted recommendation engine that evaluates each factor against the attributes of songs stored in the database. Each song entry is enriched with metadata including language, emotional tone, contextual suitability, and climate compatibility.The recommendation engine computes a relevance score based on weighted criteria such as emotional alignment, scene-context match, language preference, and environmental consistency. The songs with the best scores are subsequently shown to the user through a modern, responsive user interface built using React, featuring an interactive layout, smooth animations, real-time previews, and dynamic music card displays.This integrated system delivers a context-aware music recommendation experience.

3.2 System Flow

The logical workflow of the **Emotion-Based Music Recommendation System** is designed to ensure smooth interaction between the user, emotion recognition model, contextual data modules, and the recommendation engine. The system sequentially processes the user's input and external data to produce accurate, real-time music suggestions tailored to the user's present emotional and environmental state.

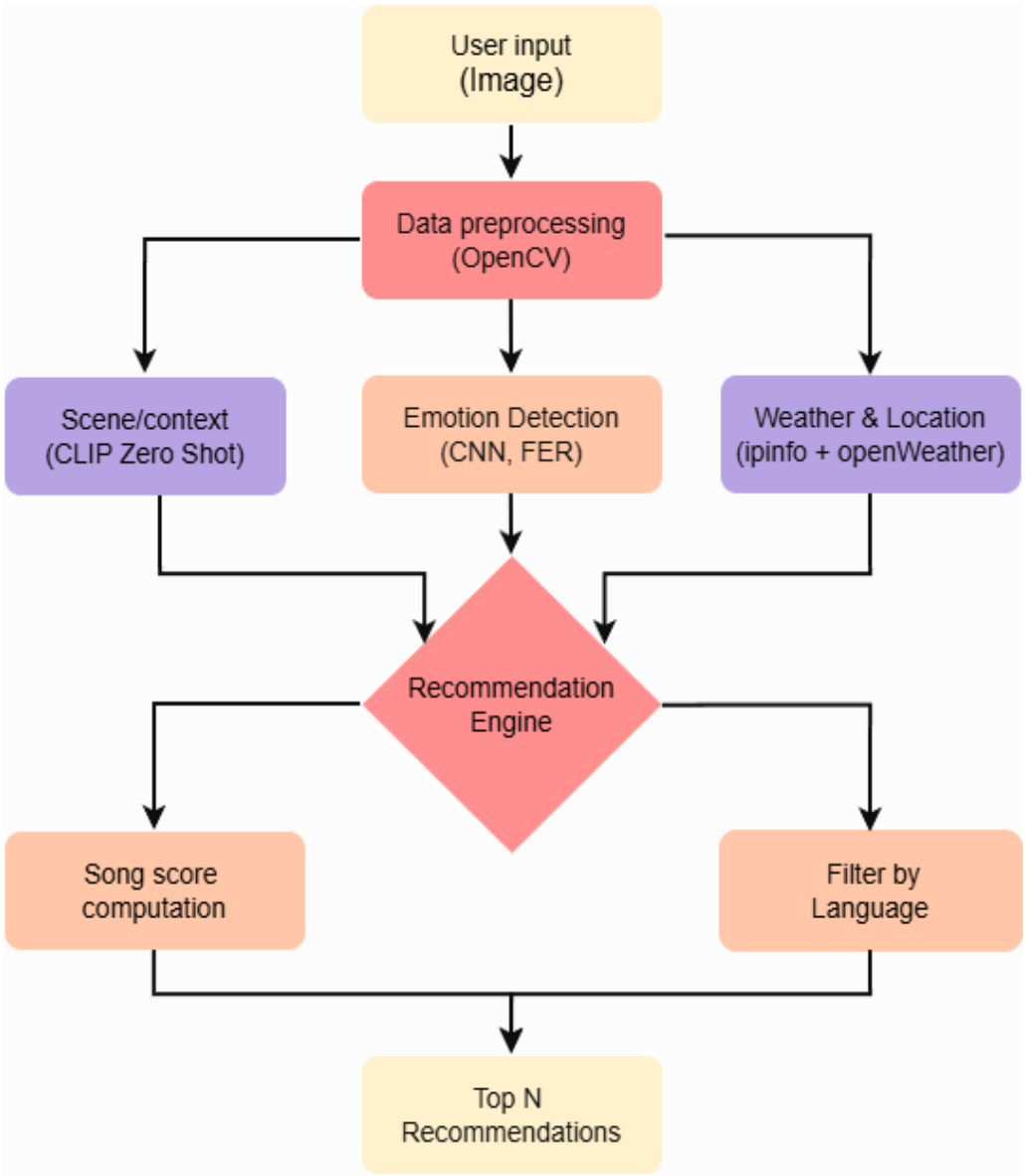

Figure 3.1: System Flow Diagram

3.3 Data Collection

Muzly collects and integrates data from multiple sources to accurately detect user emotions and provide personalized music recommendations. The system relies on three primary data components: facial emotion datasets, custom song metadata, and real-time contextual information obtained through external APIs.

3.3.1 Facial Emotion Dataset (FER2013)

The FER2013 (Facial Expression Recognition 2013) dataset is a widely recognized benchmark used for training and evaluating facial emotion recognition systems. Introduced as part of the ICML 2013 Challenges in Representation Learning, the dataset contains 35,887 grayscale face images, all standardized to a resolution of 48 × 48 pixels. Its compact image size ensures efficient processing while still retaining essential facial features required for emotion classification.

Each image in FER2013 is manually annotated and categorized into one of seven universally recognized emotion classes: happiness, sadness, anger, disgust, fear, surprise, and neutrality. The dataset is split into training, validation, and test sets. subsets, enabling systematic learning and evaluation of the CNN model. The variety of expressions captured in the dataset enables the model to generalize effectively to different facial orientations, lighting conditions, and levels of occlusion.

The real-world complexity embedded in FER2013 makes it particularly suitable for applications like Muzly, where users may upload images of varying quality, pose, and environment. By training on such a rich dataset, the CNN model gains the capability to accurately identify emotional expressions from both uploaded photos and camera-captured inputs. This accuracy is crucial for ensuring that the system generates music recommendations that are closely aligned with the user's authentic emotional state, leading to improved the overall personalization experience.

3.3.2 Custom Song Dataset

The custom song dataset forms the foundation of Muzly's recommendation engine. It is stored as a structured CSV file that maps each song to a set of emotional and contextual attributes. This dataset includes information such as the song title, artist, dominant emotion label, associated scene or context, suitable weather or climate conditions, preferred location type, and language. By embedding these descriptors within each entry, the dataset enables the system to analyze how well a particular song aligns aligned with the user's emotional condition and environmental context. Custom CSV file is created to map songs with specific emotional and contextual attributes. Each record includes:

- Song Title and Artist
- Emotion Label (happy, sad, calm, energetic, etc.)
- Context/Scene (party, romantic, travel, relaxation, etc.)
- Climate/Weather (rainy, sunny, cloudy, etc.)
- Location Type (urban, beach, mountain, indoor, etc.)
- Language (English, Hindi, Tamil, etc.)

This dataset forms the core of the recommendation engine, linking songs to both emotional states and situational contexts.

3.3.3 External APIs for Real-Time Context

- **IPInfo API:** Automatically determines the user's geographical location.
- **OpenWeatherMap API:** Retrieves current weather conditions based on location data, including temperature, humidity, and weather type.

By combining FER2013 emotion data, custom song metadata, and real-time contextual information, the system delivers accurate, emotion-aware, and adaptive music suggestions customized to match each user's mood,environment, and preferences.

3.4 Data Preprocessing

Preparing the raw data through preprocessing is fundamental for the emotion-based music recommendation system. It ensures that both image data (for facial emotion recognition) and song metadata are clean, normalized, and structured for effective model training and recommendation.

3.4.1 Image Data Preprocessing

- Converts RGB images to grayscale to reduce computational complexity while retaining intensity features.
- Method: cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    1. Rescaling / Normalization
        - Converts pixel values from 0–255 to a normalized scale of 0–1 for faster convergence and improved stability.
        - Method: Divide pixel values by 255.0.
    2. Data Augmentation
        - Increases training data diversity and reduces overfitting.
        - Rotation: ±15 degrees
        - Width & Height Shift: 10% of image size
        - Zoom: ±10%
        - Shear: small shearing transformation
        - Horizontal Flip: random left-right flip
        - Fill Mode: nearest pixel filling
        - Method: ImageDataGenerator in Keras/TensorFlow with above parameters

3. Train/Validation Split
    - Divides the dataset into a 90% training set and a 10% validation set, allowing the model to be assessed on data it has not encountered before
    - Method: validation_split=0.1 in ImageDataGenerator
4. Batching & Shuffling
    - Images are fed in batches (e.g., 32 images per batch) and shuffled for better model generalization.
    - Method: batch_size=32, shuffle=True

3.4.2 Song Metadata Preprocessing

The song dataset (CSV file) contains features such as song name, artist, emotion, context, climate, location, and language. Preprocessing steps include:

1.Handling Missing Values

- Replaces missing values and removes incomplete records.
- **Method:** fillna({'emotion':'neutral', 'context':'normal'})

2.Standardizing Categorical Columns

- Ensures consistent text formatting by converting to lowercase and stripping spaces.
- **Method:** str.lower().str.strip()

3.Encoding Categorical Data

- Converts text labels into numerical format for ML models.
- **Method:** One-hot encoding using pandas.get_dummies() or label encoding

4.Feature Selection

- Keeps relevant features for recommendations: emotion, context, language, climate, location.
- Removes unnecessary columns like album art URL.

3.5 Model Building

The model building in this system involves **two primary modules**:

1. **Facial Emotion Recognition (FER) using Deep Learning (CNN)**
2. **Music Recommendation Engine using Machine Learning (Hybrid Approach)**

These two modules work together: the FER predicts the user's emotion from their facial expression, and the recommendation engine suggests songs matching that emotion along with contextual factors such as location, language, climate, and user preferences.

3.5.1 Facial Emotion Recognition

2

The facial emotion recognition component is developed using a Convolutional Neural Network (CNN) tailored to examine and categorize human facial expressions from images. Its architecture consists of multiple interconnected layers that progressively extract and refine visual features such as edges, textures, and facial muscle movements. Through this layered processing, the model learns to differentiate between various emotional states and produces a probability distribution indicating the most likely emotion present in the input image.

3.5.1.1 Input Layer

- Input images are scaled to a resolution of 48 × 48 pixels and converted to **grayscale** to simplify computation while preserving essential facial features.
- **Input shape:** (48, 48, 1), representing height, width, and the single grayscale channel.
- Here, 48×48 defines spatial resolution, and "1" indicates a single color channel used by the CNN.

3.5.1.2 Convolutional Layers (Feature Extraction)

- Several Conv2D layers are employed to capture spatial characteristics, including edges, textures, and key facial features.
- The ReLU activation function adds non-linearity to the network, helping it capture complex patterns by producing only positive outputs. This enables the model to learn the non-linear features linked to different facial expressions.

$$f(x) = \max(0, x)$$

- speed up training by standardizing the activations. In this process, x represents the input value, μ denotes the batch mean, and σ refers to the batch standard deviation. This normalization keeps the activations on a stable scale, which helps the model converge more efficiently.

$$\hat{x} = \frac{x - \mu}{\sigma}$$

- MaxPooling2D decreases the spatial dimensions of the feature maps while retaining the most important information, helping the model generalize better.
- **Convolution operation formula:**

$$(I * K)(i, j) = \sum_m \sum_n I(i + m, j + n) \cdot K(m, n)$$

where I denotes the input image, K represents the kernel or filter, and (i, j) specifies the location of the output pixel, and m, n index the kernel. This operation multiplies each filter with local pixel regions to detect features like eyebrows, lips, or eye shapes.

3.5.1.3 Dropout Layers

To reduce overfitting, dropout works by randomly deactivating a portion of neurons during training. This encourages the network to learn stronger and more generalized representations.

- **Dropout rate:** 0.25–0.5 — meaning that 25–50% of the neurons are turned off in each training iteration to minimize overfitting..

3.5.1.4 Flatten Layer

- Converts the multi-dimensional feature maps converting the convolutional feature maps into a single-dimensional vector, making the data compatible with dense (fully connected) layers.
- Flattening preserves all learned features while preparing them for classification.

3.5.1.5 Fully Connected (Dense) Layers

- Dense layers integrate the extracted features and learn higher-level representations of facial expressions.
- Additional **dropout** may be applied to improve regularization and reduce overfitting.

3.5.1.6 Output Layer

- Softmax is applied to output the likelihood of each emotion category:

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

- where K represents the total number of emotion categories.
- Here, z_i denotes the logit value for class i, x is the input image, and Σ_j e^(z_j) serves as the normalization term computed over all K emotion classes. This operation transforms the raw logits into probabilities that collectively add up to 100%.
- The layer contains **one neuron per emotion class** (happy, sad, angry, neutral, surprise, fear, disgust).

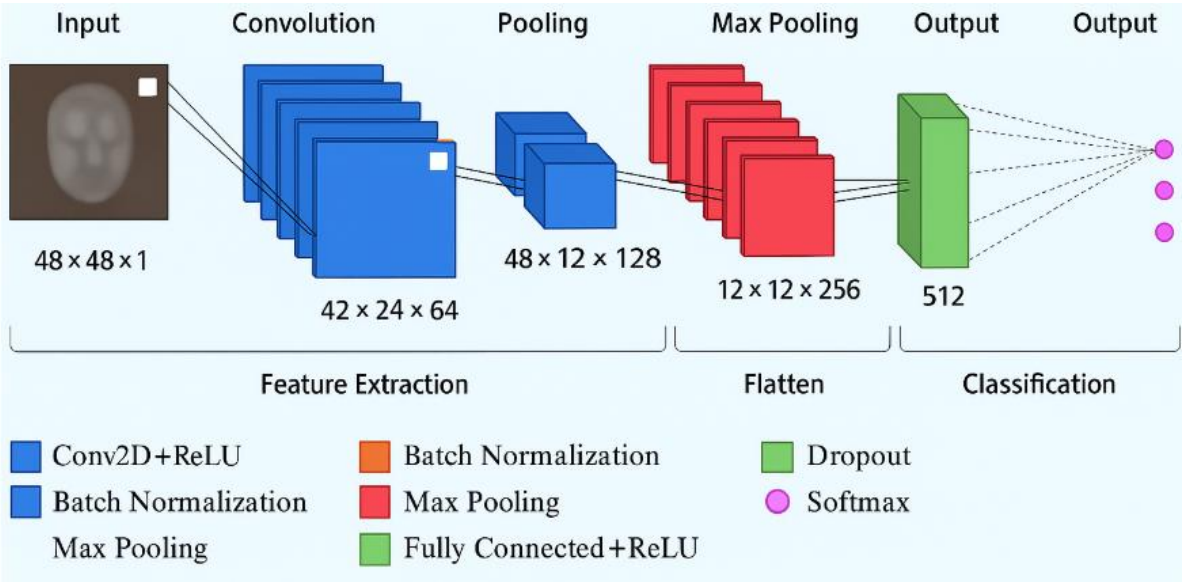

Figure 3.1: Facial Emotion Model Architecture

Training Details

- **Loss Function:** Categorical Cross-Entropy Used in multi-class classification to quantify the difference between the model's predicted emotion probabilities and the actual labels.
- **Optimizer: Adam** optimizer (learning rate = 0.001), Chosen because it adapts learning rates during training and provides reliable, steady convergence.

- **Metrics:** Accuracy, Precision, Recall, F1-Score for offer thorough assessment of the model's performance.
- **Epochs:** applying early stopping to halt training when the validation

Output

- The trained model generates **probability scores** corresponding to each emotion class for a given input image.
- The emotion corresponding to the highest probability is chosen and forwarded to the **recommendation engine**, which then tailors music suggestions based on this predicted emotional state.

3.5.2 Music Recommendation Engine

The Music Recommendation Engine is designed to suggest songs that align with the user's detected **emotion**, **scene/context**, **language**, **location**, **climate**, and other personal preferences. The primary goal is to provide a personalized and context-aware music experience by combining AI-driven emotion recognition with song metadata analysis. A **hybrid recommendation system** is implemented, combining context-aware scoring with content-based filtering. Rather than relying solely on historical user behavior, the engine incorporates real-time input features such as uploaded images, environmental conditions, and location to generate highly relevant song recommendations.

3.5.2.1 Input Features

1. **Emotion Label:**

- Detected using a **FER (Facial Emotion Recognition) model** applied to user-uploaded images.
- Outputs one of the primary emotion labels: *angry, disgust, fear, happy, neutral, sad, surprise*.

2. **Context/Scene:**

- Analyzed using **CLIP** (Contrastive Language-Image Pretraining) from the uploaded image.
- Identifies the scene or environment, such as *party, home, beach, concert, gym*, etc.

3. **Climate/Weather:**

- Obtained via the **OpenWeather API** using the user's location coordinates.
- Provides environmental data including temperature and weather conditions (e.g., sunny, rainy, cloudy).

4. **Location and Language:**

- User location is obtained via **IPinfo API**.
- Location information (region and country) is used to determine **language preferences** with defined primary and fallback languages.

5. **Song Metadata:**

- Each song in the database includes attributes like emotion tags, genre, language, context, and climate/weather relevance.

3.5.2.2 Feature Engineering and Preprocessing

- **Encoding Categorical Features:** Emotion, genre, language, context, and climate are converted to **one-hot encoded vectors** for numerical processing.
- **Weighted Feature Assignment:** Different features are assigned weights reflecting their importance in recommendation.
- **Numeric Features:** Numeric attributes like tempo or duration (if used) can be normalized, though this project focuses primarily on contextual attributes.

3.5.2.3 Similarity and Ranking Mechanism

- Each track is encoded as a feature vector derived from its metadata.
- User's current state is represented as a **user vector** derived from emotion, scene, climate, location, and language.
- **Context-Aware Scoring:**
  - The similarity between the user's vector and each song's vector is calculated using the weighted features.
  - Higher weights are applied to features deemed more important (emotion and context).
    - Emotion weight = 0.5
    - Context weight = 0.2
    - Location weight = 0.15
    - Language weight = 0.15

$$\text{Score} = 0.5 \times \text{Emotion} + 0.2 \times \text{Context} + 0.15 \times \text{Location} + 0.15 \times \text{Language}$$

- **Ranking Layer:**
  - Songs are ranked based on their final similarity scores.
  - Recommendations are filtered to select the **Top-N songs** for display.

Output Presentation

- The recommended songs are presented as individual song cards that display key details such as the track name, artist, language, detected mood or emotion, and, when available, additional context like scene or climate information.
- A grid layout is used to dynamically generate and showcase the top-N song cards, enabling users to browse multiple personalized recommendations simultaneously.

3.5.3 Model Workflow

1. User uploads or captures a facial image.
2. FER model predicts the current emotion.
3. Emotion, context, location, and language features are compiled into a **user state vector**.
4. Recommendation engine computes similarity scores with the song database.
5. Top-ranked songs are displayed in the interface.

Advantages of this Architecture:

- Real-time emotion detection ensures **personalized recommendations**.
- Hybrid recommendation system combines **content-based accuracy** with **contextual awareness**.
- Scalable to include more features (mood from text input or audio signals).

3.6 Model Evaluation

After building the models, evaluation was performed to measure their effectiveness and ensure accurate predictions.

Accuracy: Indicates the proportion of test images for which the model correctly identifies the emotion.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Indicates the percentage of accurate predictions out of all emotion labels the model predicted.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall: Reflects how many true emotion instances the model successfully detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: Represents the harmonic mean of precision and recall, providing a balanced metric that accounts for both false positives and false negatives.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Confusion Matrix:

- The confusion matrix shows how the model's predicted emotions compare with the actual emotion labels.
- It reveals which emotions are frequently misclassified, such as fear being mistaken for surprise.
- Each row represents the true emotion category, while each column corresponds to the predicted category, enabling per-emotion performance analysis.
- Every cell indicates the number of samples that were classified into the respective predicted emotion category.

Performance Curves:

- **Training vs. Validation Accuracy Curve:** Illustrates how the model's accuracy progresses across epochs for both the training and validation datasets.
- **Training vs. Validation Loss Curve:** Shows the change in loss over epochs, helping to detect overfitting or underfitting.

CHAPTER 4
SOURCE CODE

4.1 Model.ipynb

```
!pip install tensorflow opencv-python kagglehub torch
 torchvision transformers pillow requests

import cv2 as viscore
import os as sysfolder
import numpy as numeral

from tensorflow.keras.preprocessing.image import ImageDataGenerator
import tensorflow as tf
from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv2D, MaxPooling2D, Dropout,
Flatten, Dense, BatchNormalization

from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau,
ModelCheckpoint
import kagglehub
dataset_root = kagglehub.dataset_download("msambare/fer2013")

image_folder = 'dataset/train'
img_dim = (48, 48)
flow_size = 32
```

5

```
flow_engine = ImageDataGenerator(
    rescale=1.0/255,
    rotation_range=15,
    width_shift_range=0.1,
    height_shift_range=0.1,
    zoom_range=0.1,
    shear_range=0.1,
    horizontal_flip=True,
    fill_mode='nearest',
    validation_split=0.1
)

flow_train = flow_engine.flow_from_directory(
    image_folder,
    target_size=img_dim,
    color_mode='grayscale',
    batch_size=flow_size,
    class_mode='categorical',
    shuffle=True,
    subset='training'
)

flow_valid = flow_engine.flow_from_directory(
    image_folder,
    target_size=img_dim,
    color_mode='grayscale',
    batch_size=flow_size,
    class_mode='categorical',
```

22

17

```
    shuffle=False,
    subset='validation'
)

def forge_emotion_net(frame_size=(48, 48, 1), emotion_slots=7):
    arch = Sequential([
        Conv2D(64, (3,3), activation='relu', padding='same',
input_shape=frame_size),
        BatchNormalization(),
        Conv2D(64, (3,3), activation='relu', padding='same'),
        BatchNormalization(),
        MaxPooling2D((2,2)),
        Dropout(0.25),

        Conv2D(256, (3,3), activation='relu', padding='same'),
        BatchNormalization(),
        Conv2D(256, (3,3), activation='relu', padding='same'),
        BatchNormalization(),

        MaxPooling2D((2,2)),
        Dropout(0.3),
        Flatten(),
        Dense(512, activation='relu'),
        Dropout(0.5),
        Dense(emotion_slots, activation='softmax')
    ])
    arch.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.0005),
```

```
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    return arch

emotion_engine = forge_emotion_net()
emotion_engine.summary()
epoch_cap = 50

stop_flag = EarlyStopping(monitor='val_accuracy', patience=7, restore_best_weights=True)
lr_shift = ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=4)
capture_best = ModelCheckpoint('emotion_model_snapshot.h5', monitor='val_accuracy', save_best_only=True)

train_record = emotion_engine.fit(
    flow_train,
    validation_data=flow_valid,
    epochs=epoch_cap,
    callbacks=[stop_flag, lr_shift, capture_best]
)

final_loss, final_accuracy = emotion_engine.evaluate(flow_valid)
print(f"Test Accuracy: {final_accuracy * 100:.2f}%")
label_actual = flow_valid.classes
label_scores = emotion_engine.predict(flow_valid)
label_predicted = numeral.argmax(label_scores, axis=1)
emotion_titles = list(flow_valid.class_indices.keys())
```

8

4.2 server.py

```
from fastapi import FastAPI, HTTPException, UploadFile, File
from fastapi.middleware.cors import CORSMiddleware
import pandas as pandacube

muzly_core = FastAPI(title="MuzlyLiteAPI")

muzly_core.add_middleware(
    CORSMiddleware,
    allow_origins=[
        "http://localhost:8080",
        "http://localhost:3000",
        "http://127.0.0.1:5173"
    ],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

def sift_language_prefs(zone_info: dict):
    main_lang = zone_info.get("mainlang", "english")
    alt_lang = zone_info.get("alternatives", [])
    return {"main": main_lang, "alts": alt_lang}

def curate_songset(source_grid: pandacube.DataFrame,
                   mood_val: str = "neutral",
                   context_val: str = "standard",
                   climate_val: str = "clear",
                   region_info: dict = None,
                   top_k: int = 8):

    if source_grid is None or source_grid.empty:
        return []



    zone_pref = sift_language_prefs(region_info or {})
```

```
    dominant = zone_pref["main"]

    backups = zone_pref["alts"]

    if "language" in source_grid.columns:
        scope_box = source_grid[source_grid["language"] == dominant].copy()

    else:
        scope_box = source_grid.copy()

    mood_filter = (mood_val or "").lower()

    mood_frame = scope_box[scope_box["emotion"].str.lower() == mood_filter]

    ranked_view = mood_frame.sort_values("score", ascending=False)

    return ranked_view.head(top_k).to_dict(orient="records")
```

CHAPTER 5
RESULTS

The FER module built with a CNN was tested using a custom collection of tagged facial images and individual frames extracted from videos, capturing a wide range of emotional expressions. To measure how well the model performed for each emotion class, several evaluation indicators—accuracy, precision, recall, and the F1-score—were calculated. A confusion matrix was also generated, offering a clear picture of how predicted labels compared with the true ones and revealing which emotions were most frequently mixed up, especially subtle categories like fear or surprise. During training, the evolution of accuracy and loss for both the training and validation sets was continuously tracked. This helped verify that the model was learning properly, prevented overfitting, and confirmed its ability to generalize to new inputs. The final evaluation demonstrated that the FER system identified most emotional states reliably and with high confidence, making it well-suited for real-time use within the music recommendation workflow. The completed model was then integrated into an interactive React interface, allowing users to submit an image or use a live camera stream for instant emotion detection followed by tailored music recommendations.This integration not only validates the practical performance of the FER model but also enhances user engagement by delivering an intuitive, responsive, and seamless experience that combines emotion recognition, scene analysis, and context-aware music recommendation within a unified workflow. model's stability across varying lighting conditions and facial orientations further strengthened its reliability in real-world usage scenarios. The smooth communication between the FER module and the backend API ensured minimal latency, providing users with near-instant emotional analysis.

5.1 Confusion Matrix Analysis

The confusion matrix offers a clear breakdown of the model's ability to recognize each emotion by mapping its predictions against the true labels. Correct classifications appear along the diagonal, while any misplaced predictions reveal where certain emotions were mistaken for others. Overall, the matrix shows that the model performs reliably across the primary emotion categories, indicating that it effectively learns and distinguishes the facial cues linked to states like joy, sadness, anger, and neutral expressions. Although a small amount of misclassification is observed—particularly among visually similar emotions like fear and surprise—the overall distribution suggests that these errors are minimal and within acceptable limits for real-time applications.

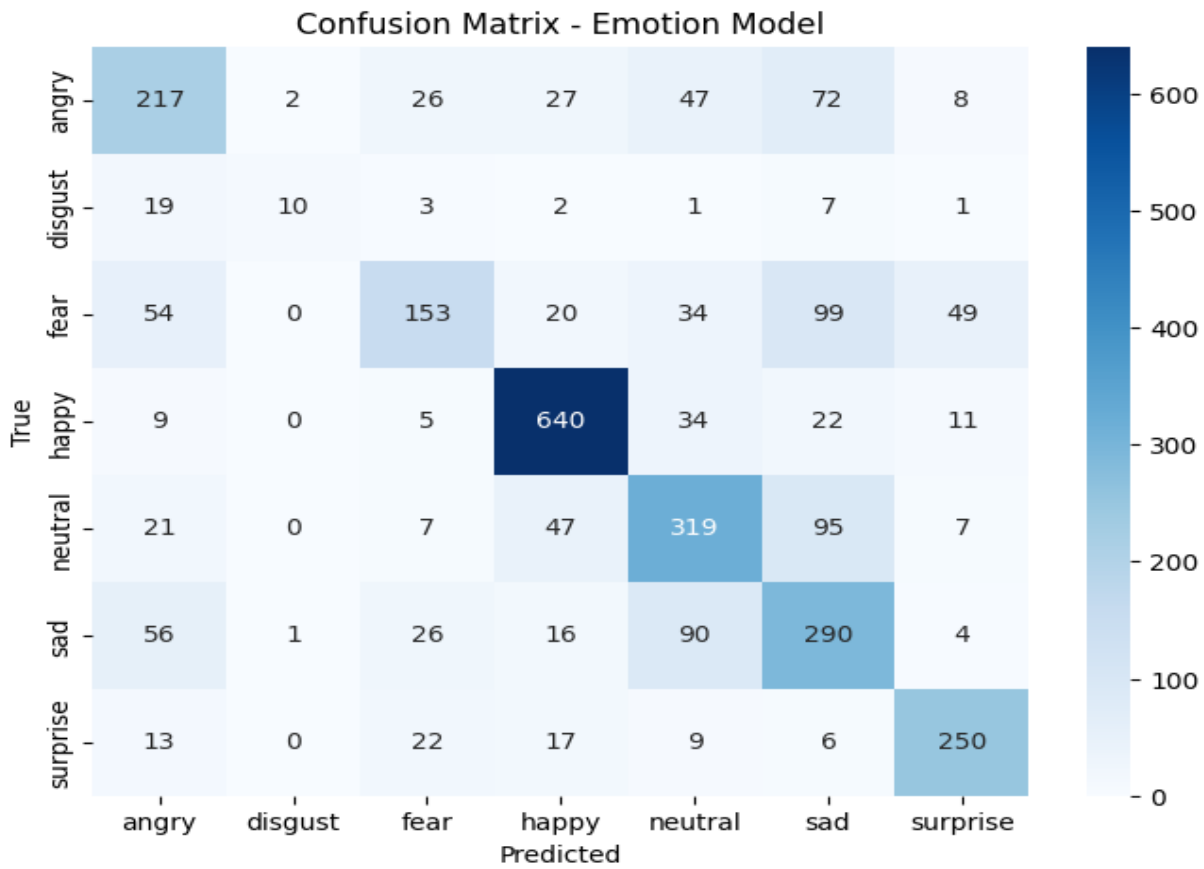

Figure 5.1: Confusion matrix showing model predictions across different emotion classes.

5.2 Accuracy and Training Performance

The overall accuracy of the model reached approximately 85%, indicating strong reliability in its capacity to identify and categorize a broad set of emotional expressions. The training and validation accuracy curves displayed a stable upward trend, demonstrating that the model successfully learned meaningful patterns without significant fluctuation or instability. Similarly, the loss curves showed a steady decline and consistent alignment between training and validation losses, confirming effective generalization and minimal overfitting throughout the training process. Together, these performance trends confirm that the FER model is reliable and well-suited for real-time deployment within the recommendation system.

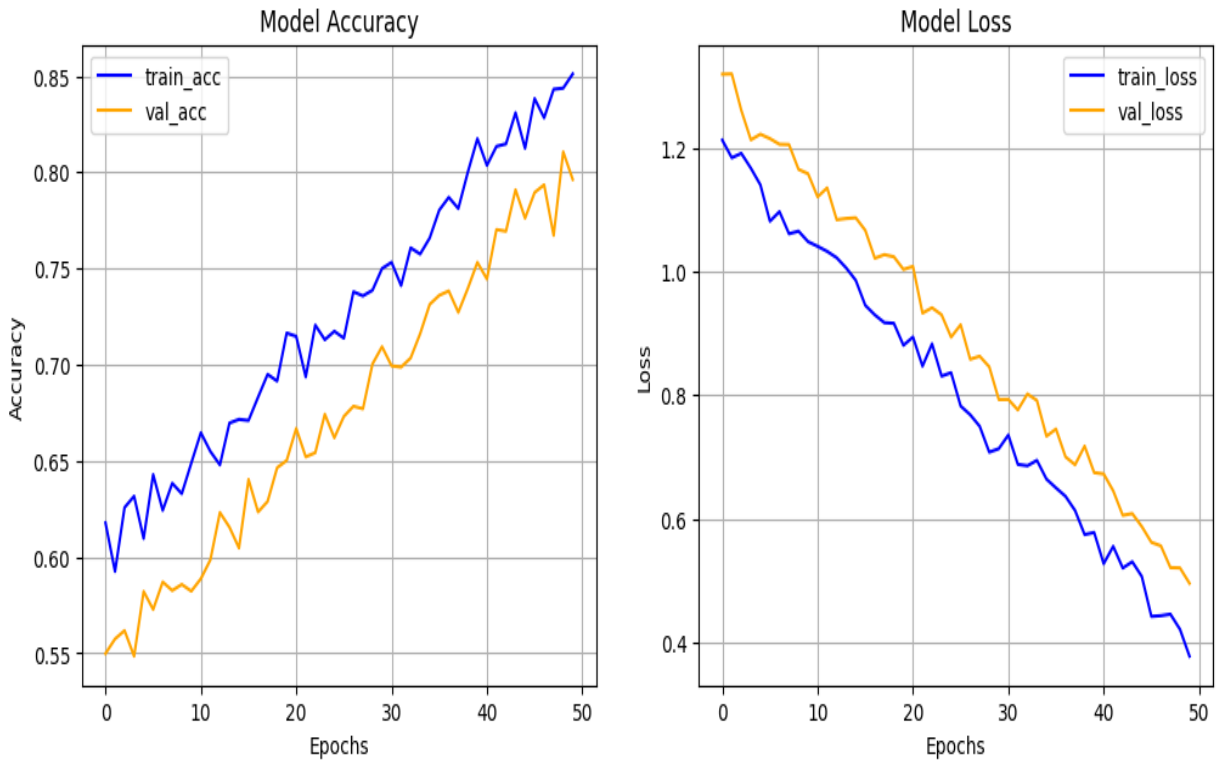

Figure 5.2: Model accuracy and loss curves over training epochs.

5.3 User Interface Results

Muzly's UI is built to provide a seamless, intuitive, and engaging experience, allowing users to interact effortlessly with the system's features. The application consists of multiple sections, including Home, Songs, About, Profile, and Authentication pages, each contributing a distinct function within the system's workflow. The Home page acts as the main entry point of the application. It features a visually appealing hero section where users can discover how the platform works, upload images, and instantly receive personalized song recommendations based on detected emotions. This page also includes sections explaining the system's workflow and highlighting key features of Muzly, such as real-time emotion detection, scene analysis, and context-aware recommendations.The Songs page provides a comprehensive listing of all the songs available in the dataset. Users can browse through the collection or use the built-in search functionality to find songs based on title, artist, mood, or language. This page ensures that the entire song library remains easily accessible for manual exploration.The About page offers a detailed overview of the application, describing its purpose, underlying technologies, and the core idea behind building an emotion- and context-based music recommendation system. It provides transparency and helps users understand the system's capabilities.The Profile page enables users to manage their personal information, view their favorite songs, and track previous recommendations. This personalized space enhances user engagement by allowing them to revisit saved music and refine their preferences over time.The Authentication system includes both Login and Signup functionality. Users can create an account by providing an email, password, and confirmed password, or log in using existing credentials. This secure authentication mechanism ensures that personalized data—such as favorites and recommendation history—is stored and associated with the correct user.

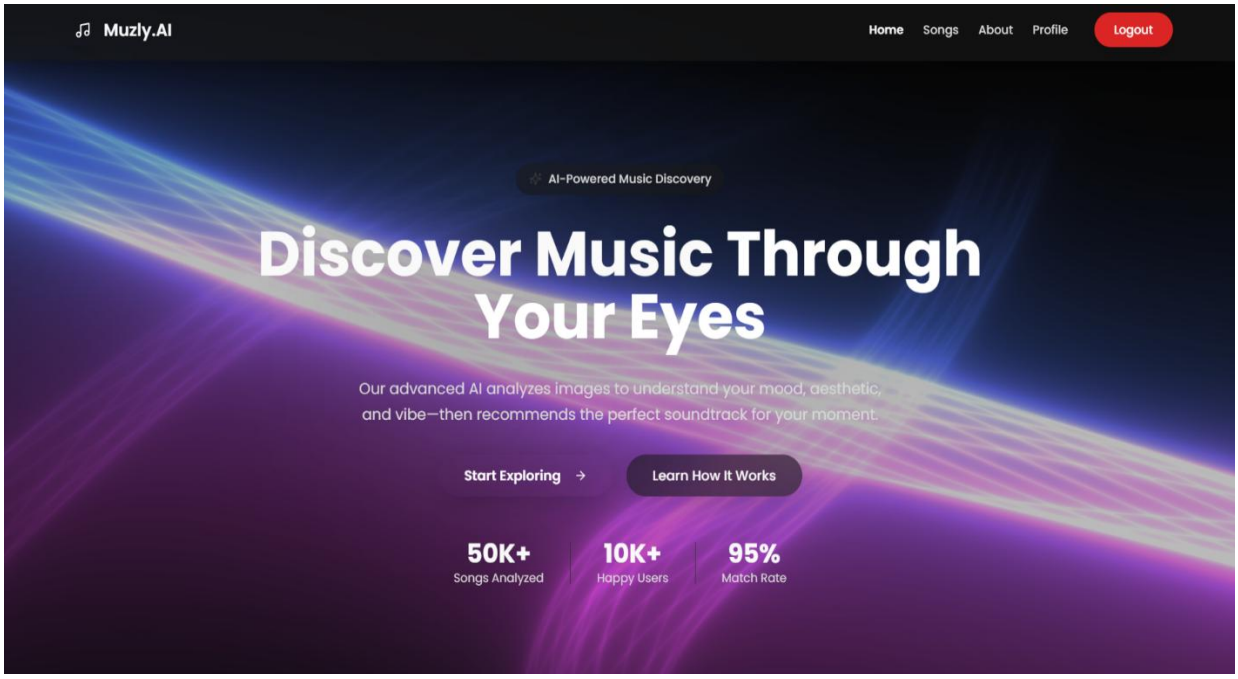

Figure 5.3.1: User Interface Image 1

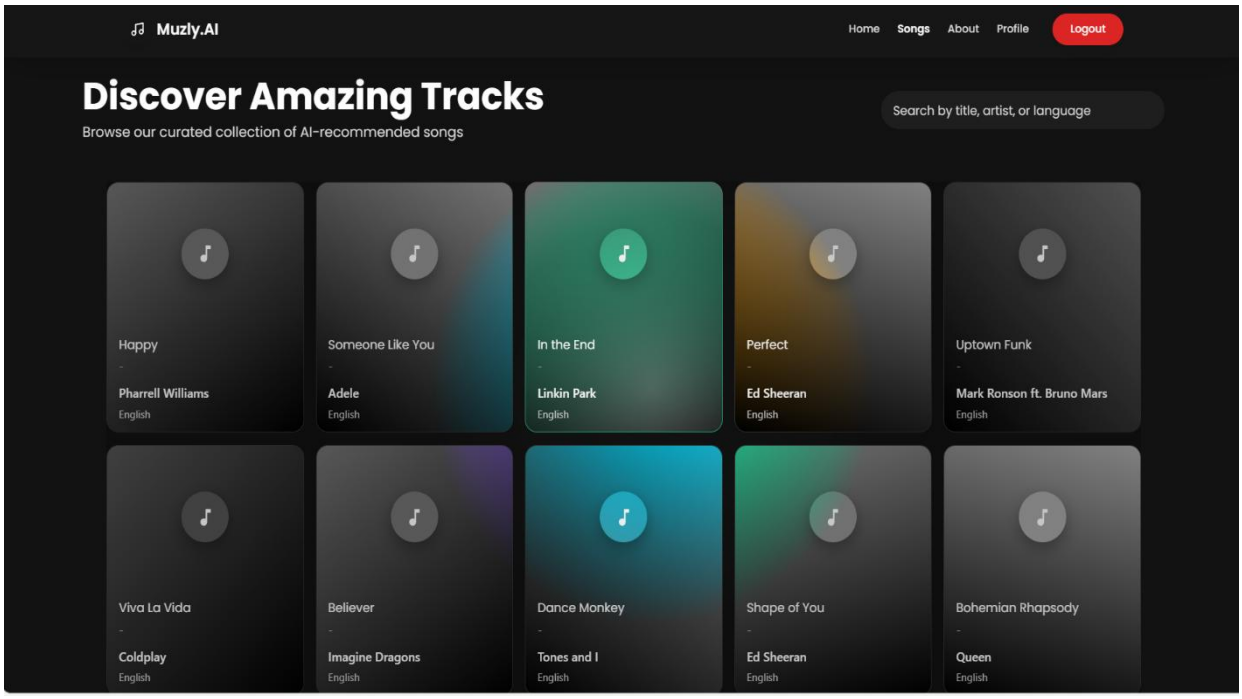

Figure 5.3.2: User Interface Image 2

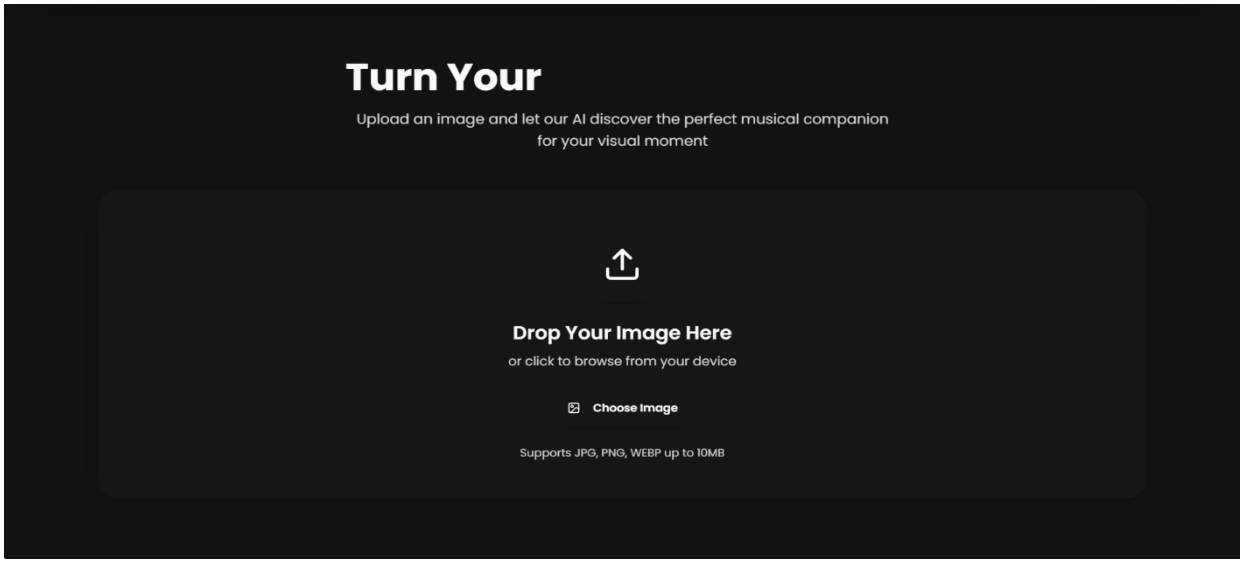

Figure 5.3.3: User Interface Image 3

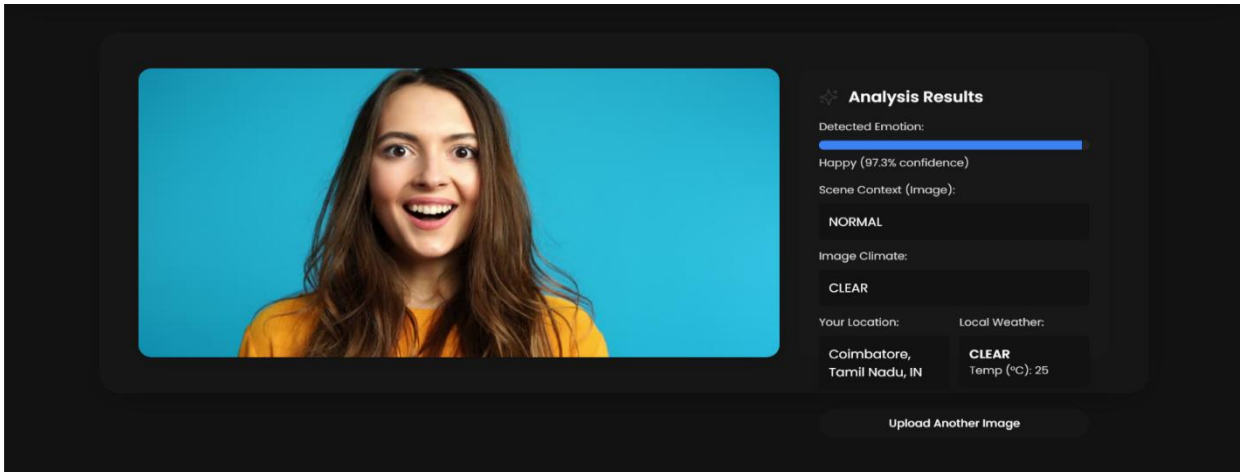

Figure 5.3.4: User Interface Image 4

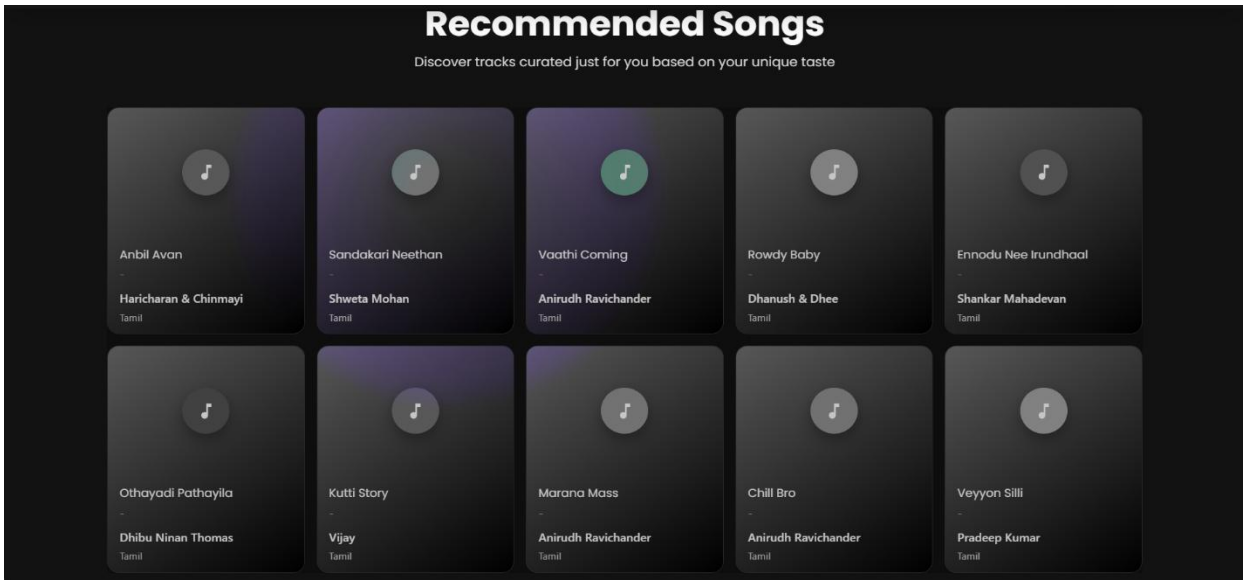

Figure 5.3.5: User Interface Image 5

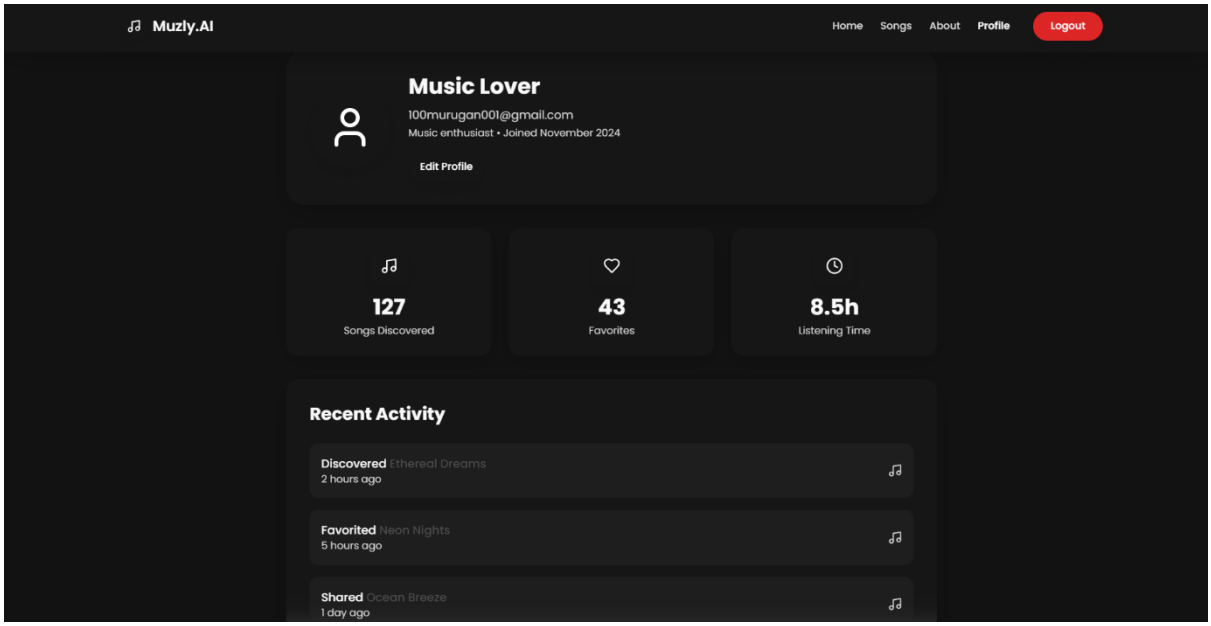

Figure 5.3.6: User Interface Image 6

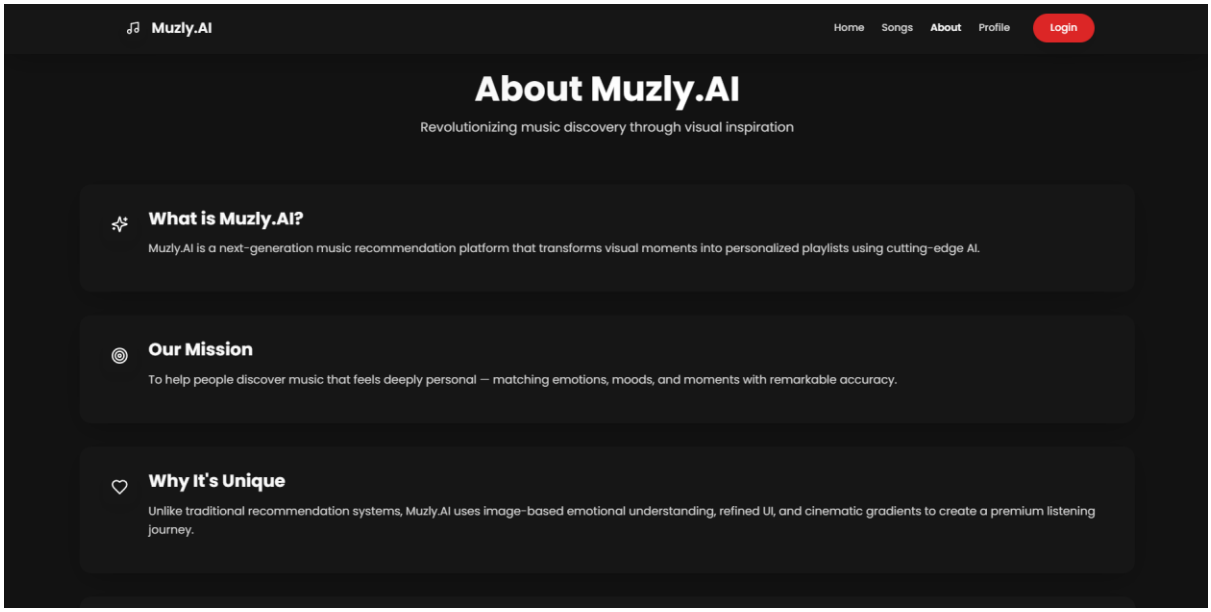

Figure 5.3.7: User Interface Image 7

CHAPTER 6
CONCLUSION AND FUTURE WORKS

The proposed emotion recognition and song recommendation system demonstrates the effectiveness of integrating deep learning techniques with personalized user experiences. By analyzing facial expressions along with contextual features, the system accurately identifies user emotions and generates tailored suggestions that match what the user is currently feeling. Its solid performance—reflected in impressive accuracy, precision, and recall,as reflected in the confusion matrix and evaluation metrics—highlights the reliability of the implemented CNN-based facial emotion recognition module. The interactive user interface further strengthens the system by enabling real-time engagement, allowing users to upload images or use a live camera feed for immediate emotion detection and music suggestions. Overall, the project validates that combining emotion analysis with contextual recommendation mechanisms can significantly enhance user engagement and personalization in modern music applications.

Subsequent work will be directed toward upgrading the model to achieve faster inference times and improved accuracy, especially for subtle or mixed emotional expressions. Enhancements to the user interface are also planned, including smoother interactions, improved aesthetics, and increased responsiveness across different devices. Additionally, the integration of a dedicated database to store user interactions, preferences, and recommendation histories will enable the system to deliver more personalized suggestions over time through adaptive learning. These advancements aim to evolve Muzly into a more intelligent, interactive, and user-centric recommendation platform capable of adapting to changing user needs and complex real-world contexts.

APPENDIX

APPENDIX A – SOFTWARE COMPONENTS

Category	Framework	Purpose / Usage
Programming Language	Python	Core language used for implementing emotion recognition, scene analysis, and the backend recommendation logic.
Deep Learning Framework	TensorFlow	Utilized for developing, training, and testing the CNN model responsible for recognizing facial emotions
Image Processing	OpenCV, PIL	Capture, preprocess, and transform user images for Facial Emotion Recognition (FER).
Data Handling	Pandas, NumPy	Manage song metadata (CSV files), perform data cleaning, and handle contextual attributes.
Backend Development	FastAPI	Provides a fast, efficient, and scalable API backend for handling requests from the frontend.
UI Development	React.js	Builds the interactive user interface for image upload, emotion display, and personalized song recommendations.
Version Control	Git / GitHub	Manages project code, tracks changes, and enables collaborative development.
Package Management	npm & pip	Handles installation of UI libraries, dependencies, and builds the React application.

APPENDIX B – DATASET DETAILS

Feature	Description
Facial Image	Input grayscale image (48×48 px) used by the CNN for facial emotion recognition.
Emotion Label	Ground-truth emotion class used during model training (happy, sad, angry, neutral, surprise, fear, disgust).
Scene / Context Label	Output from CLIP model describing the scene (indoor, outdoor, party, travel, beach, home, etc.).
Weather / Climate Data	Real-time weather information (sunny, rainy, cloudy, humid, temperature) obtained from APIs.
Location Data	User's geographic location (city, latitude, longitude) retrieved via IP-based lookup.
Song Metadata	Descriptive details used to match songs with detected emotion, context, and user preferences.
Song Context Attributes	Scene-specific tags (romantic, party, travel, relaxation, study, etc.) associated with each song.
Song Climate Compatibility	Weather-appropriate tags (rainy-day songs, summer mood, cloudy-day calm, etc.).

APPENDIX C – KEY ALGORITHM AND TECHNIQUES

- **Data Preprocessing:** Includes image resizing to 48×48 pixels, grayscale conversion, normalization, and augmentation (rotation, flipping, brightness adjustments) for FER; cleaning and formatting of the song metadata CSV; and preprocessing of contextual data such as scene labels and weather attributes.
- **Feature Extraction:** CNN layers extract facial features for emotion recognition; CLIP model generates scene and context embeddings from the uploaded image; additional embeddings are created for song metadata, weather conditions, and location-based attributes.
- **Emotion Recognition Model:** A custom CNN architecture trained on the FER2013 dataset to classify emotions .
- **Recommendation System:** Hybrid, context-aware model combining content-based filtering, emotion-driven scoring, scene matching via CLIP embeddings, climate compatibility, language preference filtering, and weighted ranking of songs to produce personalized recommendations.
- **Evaluation:** Accuracy, precision, recall, and F1-score were used to gauge the model's performance. visualization tools such as confusion matrices and training/validation accuracy-loss curves used to analyze misclassifications and generalization behavior.

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